

FITFLOW FITNESS TRACKER

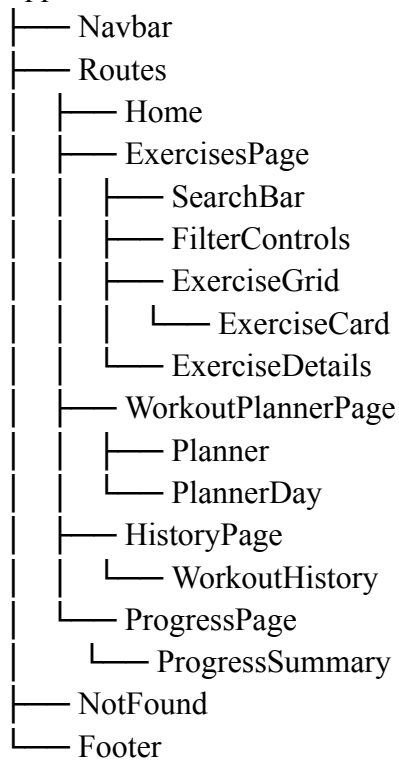
Project Planning Document

Project: React Fitness Tracker & Workout Planner

Technology: React, JavaScript, CSS, Jest, React Testing Library

1. COMPONENT HIERARCHY

App



FitnessProvider



2. DATA FLOW DIAGRAM

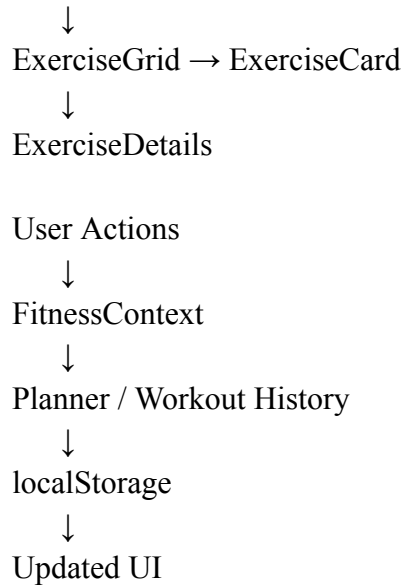
Exercise Data



ExercisesPage



Search / Filters



Exercise information is stored in the application data. User-generated planner and workout-history data is managed through FitnessContext and persisted using localStorage.

3. COMPONENTS TO BE CREATED

App, Navbar, Footer, Home, ExercisesPage, SearchBar, FilterControls, ExerciseGrid, ExerciseCard, ExerciseDetails, WorkoutPlannerPage, Planner, PlannerDay, HistoryPage, WorkoutHistory, ProgressPage, ProgressSummary, NotFound and FitnessProvider.

4. PROPS FLOW

ExercisesPage passes filtered exercises to ExerciseGrid. ExerciseGrid passes individual exercise objects to ExerciseCard. Exercise information includes properties such as name, category, muscle, difficulty and image.

Planner passes the selected day and exercises to PlannerDay. Shared planner and history functions are provided through FitnessContext.

5. STATE MANAGEMENT STRATEGY

useState manages temporary component state such as search terms, filters and selected exercises. FitnessContext and useContext manage shared planner and workout-history state. localStorage persists user data between sessions without requiring a backend. React Router manages page navigation.

6. TESTING STRATEGY

UI Components: Jest and React Testing Library will test rendering, props, buttons, links and user interactions.

Pages: Tests will verify required content, child components and important navigation.

FitnessContext: Tests will cover initial state, localStorage loading and saving, adding/removing exercises, duplicate prevention and workout history.

Utility Functions: Tests will verify expected results, edge cases, filtering and sorting.

Integration: Key user flows will be tested, including searching for an exercise, adding it to the planner, logging a workout and viewing workout history.

Test coverage will be monitored using Jest's --coverage option.